

00:00

Hello everyone. This is our first lecture on Introduction to Data Science and AI, SIT112. This is the first part of lecture week one. There's going to be a part two. In part one, we're going to talk about the unit outline. My name is Davut. I'm a senior lecturer in software engineering at the School of Information Technology at Deakin University. I'm also the unit chair for this unit, as well as the lecturer. So this is the overview of the unit. We have 11 weeks of teaching that includes lectures and workshops and of course working on the tasks. You are expected to complete all your tasks and include them in your learning portfolio and submit it by the deadline that is the end of week 12. During the university close dates there is no lecture or workshop. For the workshops you can attend any other workshops during the week that suits you on campus or online. And for any lectures affected by the university close dates there would be a recording made available online.

01:26

Here you can see an overview of the tasks that you need to work on during this unit. You can see each task has a start date, due date and deadline. These tasks are all listed on OnTrack so you can view them from there, check their details. The start date is the date that we expect you to start working on each task and if you want to start earlier that's fine. All tasks are available right now on OnTrack so you can start working on them anytime you like but the markers will only start marking all the tasks or the submitted tasks after their start date. So if you submit too early, you will have to wait until the start of the task to get any feedback. Regarding the due date, due date is essentially to encourage you to submit your task as early as possible. That's the expected date. If you couldn't make it, that's fine. You don't need to apply for any extensions. You can just continue working on the tasks and include them in your final portfolio by the end of week 12. That's Friday of week 12. And that's the deadline. Another point about the due date is that only if you submit your task by the due date you can expect feedback from the markers. Markers are very busy marking and you cannot keep going back and forth between tasks so if you want timely feedback from the markers make sure to submit your task by the due date. Will you receive any feedback if you submit your task after your due date? Markers are not expected to provide you any feedback on OnTrack, but you can still go to the workshops and talk to your tutor a



nd they're going to give you feedback on the status of your tasks and help you to complete them.

03:36

All right, so how to access the lectures. The lectures will be online and they can be accessed from the link that's provided on the CloudDeakin site. You need to go to this link, online classrooms, and under online classrooms there will be a link to the online lectures. And of course, contents of the lectures. So each lecture comes with some lecture notes examples which will be uploaded on the CloudDeakin site for each lecture and it can be downloaded from the CloudDeakin site. That will be under content. So you can start from content and then navigate through the links to find the contents related to each week. And the lectures will also be recorded and can be accessed from the cloud. Let's talk about the workshops. Again for online workshops for those of you enrolled as online students, those who attend online workshops, you can find the links to online workshops in their online classrooms. Others, that means on campus students, will attend the workshops on campus. Workshops are the primary venue for you to seek technical assistance, ask technical questions. That means whenever you want to show your code, get feedback on your code or any reports or any matter related to the tasks, that's the place for you to seek assistance in the workshops. You can ask for feedback from your tutors in the workshop. It means when you work on a task and you're not sure if the task is at a stage that you can consider complete, you can ask your tutor to help you find that out. Also if you get stuck anywhere and you need some help to move forward, you can ask for help. That's perfectly fine. The tutors will walk you through the tasks during the workshops and help you complete them, but this help is limited depending on the type of the task.

06:07

We have three types of tasks, pass tasks, credit tasks, and distinction, high distinction. So the pass tasks, these are fully guided tasks. So you have been given not only the questions, but also the solutions. And you are essentially asked to understand the solutions and do it yourself and then reflect on your understanding of the solutions and the tasks. Then you have credit tasks that are still guided but there's limited guidance available and you can of course ask for assistance from your tutors on all the tasks. And for distinction, high distinction tasks, we expect you to complete them independently. However, that doesn't mean that you cannot get help or seek help during the



workshops. That's perfectly fine to ask questions and get help during the workshops. The workshops are structured as follows. The tutor will introduce you to the contents of the workshop, might quickly recap on the contents of the previous week and help you get started. And you will start working on the tasks and the tutor might come to you and explain to you how to complete the tasks or explain to you the solutions when appropriate. That mainly concerns the pass tasks where you have also been given the solutions. Then you'll continue working on the tasks and ask questions when you need help. So as I mentioned, workshops are the primary place for you to seek help and to get assistance on the tasks and to understand different aspects of the unit. So if you have any questions, you should be feeling comfortable to discuss and ask in the workshops. If you're attending online workshops and you wish to have a one-to-one discussion with a tutor, you can always ask them to arrange a one-to-one session for you so they could create a breakout room and you can talk to them in private. If you want to, for example, share your screen or show them code and ask specific questions, feel free to do that. And you can ask your tutor and they will assist you.

08:36

Okay, do's and don'ts of the workshops. Please avoid emailing code or screenshots of your code to the teaching team outside workshop hours. Instead, demonstrate your solutions, that is, the code or your reports to the tutors during the workshops. There's no point in sending your code or screenshots of your code to tutors because they're not working outside workshop hours and they won't be able to compile and run your code and they might be able to run it at their side but still it doesn't work on your computer. Sometimes it's not easy to find the cause of the error by just looking at the screenshot. So there will be lots of issues in resolving technical problems through email. So the best way to resolve these issues is to attend your workshops, share code on your device to the tutor and they'll sit next to you and they will walk you through the code and fix the issue together. And this is how we can learn. Also, if you're attending online workshops, you can, of course, share your screen, show your code to the tutor and again, resolve the issue together. The tutors can help you fix your code in a one-to-one discussion, as I mentioned, and they can also show you how to use external resources, such as ChatGPT, to fix your code while improving your programming or problem-solving skills. So we encourage everyone to use ChatGPT to improve their understanding on the topics and to get help on completing tasks when appropriate. The third point here is bring your own device. That is a Windows, Mac, Linux la



ptop. Do not use tablets or Chromebooks in the workshops. Bring your own device and make sure your device is ready for the workshop. Some of the classrooms might have limitations in terms of the power plugs or things like that, so make sure your device is properly prepared for the workshop.

11:07

Okay, let's dive into the interesting part of the unit, the assessments. So assessments in this unit are carried out through completing a set of tasks. You'll be working on pass, credit, distinction, and high distinction tasks with different objectives. The pass tasks are intended to cover your basic understanding of the unit and they're fully guided, which means you'll be given not only the questions, but also the solutions. You're expected to understand the solutions and do it yourself and then reflect on your understanding. For the credit tasks, however, you're not given the solutions but you're given, in a way, partial solutions. So you can use the contents that have been provided to you in the question and the instructions that have been provided to you to develop the full solutions. And of course, the tutors will be there to help you. This is an area that you can start utilizing tools such as ChatGPT to assist you in completing your tasks when especially you need some help with your programming skills. Distinction tasks, there's only one distinction task and there's also one high distinction task. In these tasks you're going to work on some real world problem and you're expected to complete them independently. However, as I mentioned you can still use tools such as ChatGPT and you can also get some support from your tutors to work on the distinction, high distinction tasks. So in this unit, you're required to set a target grade. That's the grade that you want to achieve at the end of this unit. It's one of these grades listed here, pass, credit, distinction, high distinction. This is important to set as early as possible so you can focus only on the tasks that are required for that particular target grade. So you need to understand the minimum requirements for each target grade. For the pass grade you need to complete all the pass tasks. If you fail to complete any of the pass tasks you're going to fail. If you want to achieve a credit grade you need to meet the minimum requirement for pass grade plus complete the credit tasks. And for a distinction grade you need to meet the minimum requirement for the credit grade and complete the distinction task. That also includes preparing a video presentation of your distinction task and answering some questions about your task when required by the marker. To get a high distinction grade, you need to meet the minimum requirement for a distinction grade plus attending an interview. So an interview



w is required. The students will be asked questions about their submissions, that means the high distinction task, and they'll be required to complete some small tasks during the interviews. That includes small programming tasks.

15:07

So the tasks in this unit are managed through OnTrack. You can access the tasks, work on them, and then receive feedback on the tasks. So this screenshot shows that you can, for example, change the status of your task from not started to working on it. And then once you're ready for feedback, you can mark it as ready for feedback. If you need help, you can flag it as need help or just directly message your tutor or discuss with your tutor in the workshop. Okay, completing tasks. You're expected to complete the tasks by their due dates, but what happens if you don't complete the task by its due date? You can still continue to work on the task and complete it by the deadline, that is the end of week 12. That's the only deadline we have in this unit and that's the deadline for all the tasks. So you can still complete the task and that will be marked as part of your portfolio. But you will not get feedback on your task if it is submitted after the due date. In other words, you don't get OnTrack feedback if you submit it after the due date, but you can still go to your tutor in the workshops and present your code or your reports to them and get feedback from them in the workshop to ensure you have completed them properly. That's to ensure that we can provide timely feedback to all students, mainly for regulating the feedback process, but otherwise there's only one deadline for all the tasks, and that is the end of week 12, which means that there's no need for requesting any extension in this unit for any of the tasks because there's only one deadline and that deadline is the end of week 12, which cannot be further extended.

17:19

Alright, so before completing each task, you're also required to read the instructions in the task description as well as the completion form that comes with the task. So what are the items that you need to include in your task submissions? For your task P1, you're required to include signed assessment guidelines. So you're expected to read the assessment guidelines, sign it, and submit it on OnTrack. For Task P2 to P8, you're required to submit a task completion report. For Task C1 and C2, you're required to submit the task completion report. In addition to that, you're also required to provide your Jupyter Notebook file. That's your .ipynb file. Same applies to tasks D1 and



HD1, you also need to provide your task completion reports as well as Jupyter Notebook files. For the distinction and high distinction tasks, you're also required to provide a link to a video recording or a video presentation of your tasks so that the tutors can or the markers can access and check the details of your submission. Here's an example of a task completion report. You can see there are two important notes there. Please do not include any Python code in your report. You can however make references to different parts of the code, that means different parts of your Jupyter Notebook file when needed, but do not include any code in your task completion report. Also, please do not leave any response field empty. If it's not applicable, enter not applicable. As an example, the task completion report will provide these templates to the tasks. All right, feedback. So the feedback is listed here and it's been given to you in the assessment guideline. It's the way that feedback will probably be provided while you're on OnTrack. That will give you some ideas about where your task stands in terms of its completion. But the best way of getting feedback is actually discussing your tasks in the workshops with your tutors. So make sure you discuss your task with your tutor, show your tasks, show your code or reports to your tutors, and make sure that you've met the requirements for the tasks.

20:23

And finally, you need to submit your portfolio. That will be by the end of week 12. And you need to submit your final portfolio by the end of week 12, including all your completed tasks. And that would be the deadline, the end of week 12. As mentioned earlier, there's no need for extensions in this unit. So please do not send any requests for extensions because there's only one deadline for all tasks and that is the end of week 12. Again, please note the difference between deadline and due date. Due date is only to regulate the feedback process. So that if you submit your task after the due date of the task, you're not going to get feedback on OnTrack, but you can still get feedback in the workshop. You can still discuss your task with the tutor and they will provide feedback to you. This is just to make sure the tutors can, or the markers can provide timely feedback. So as I mentioned, there's only one deadline, that's the end of week 12, and therefore there is no need to request for extension for any of the tasks. Plagiarism, my favorite topic, not really. There's absolutely no need for plagiarism in this unit. Therefore, there's no tolerance for it. I would be really surprised if someone goes and copies somebody else's report because that's really for you to reflect on your understanding. And there's no benefit in doing such a thing when it only takes



maybe five or 10, 15 minutes to write a report or reflect on your understanding. That's for the pass tasks. For the credit tasks, there's plenty of support for you. There's ChatGPT. You can provide the instructions that are given to you to ChatGPT and get a nearly complete solution and start working on that and just fix the issues and get it to the complete state and submit. So I'll be really surprised if somebody copies another person's solution. That means zero effort and that means not even trying to understand the question. Because if you understand the question for the credit tasks, and you can translate that to ChatGPT, you can get a nearly complete solution. So that's the logic. And therefore, there's absolutely no need for plagiarism. When it comes to distinction and high distinction tasks, you are required to present your work and you are required to answer questions related to your work. So there's, again, no need for plagiarism because it would be a waste of your time because if you copy somebody else's solution and then you're asked by the tutor to answer specific questions or to do some programming tasks, for instance, during the interview if it's an HD task and you can't do that, that would just reduce your mark, your HD effort to zero. So it's a waste of your time and I'm sure that you would prefer to utilize your time more efficiently. So please do not even consider copying somebody else's work. If you need help, ask help from your tutors and ask ChatGPT. I'm sure that would be more helpful to you.

24:17

So we have organized some help sessions as additional help for your programming skills, as well as if you need some support with tasks. So feel free to utilize them. They are going to be online. They can be accessed from the CloudDeakin site. We provide the details once they're scheduled. And you can also attend multiple help sessions. So if you attended one and you needed more help you can attend another one and that's fine. Also please do not use the help sessions as a replacement for workshops. They have different intentions and they cannot replace each other. My favorite part of this half lecture, ChatGPT. We strongly recommend you to use ChatGPT in this unit appropriately. And what does it mean to use it appropriately? Of course, not easy to define, is it? But you should be able to tell. When you're using ChatGPT to learn, to improve your skills, to understand solutions, to be able to do it yourself, that's when you know that you're using it appropriately. So as long as you are using ChatGPT this way, it's perfectly fine for you to use ChatGPT for any of the tasks. And your tutors have also been instructed to help you with that. So feel free to reach out to them and get



t help if you need. What else? Here you can also see some suggestions as to how you can use ChatGPT. It's a question that I put to ChatGPT: how can I use ChatGPT to learn about the basics of data science and AI? And then you can see there's a whole list of things that you can do. So start using this powerful tool. You can almost use it as your personal tutor. So don't miss the opportunity.

26:47

All right, communication. So in this unit we have some rules for communication to ensure the quality of the communication. We start with how to ask questions. So we have two types of questions, technical questions and non-technical questions. If you have any technical questions, such as questions, for example, related to programming in Python or any questions related to the tasks, these are the kind of questions that need to be discussed during the workshops. So we use workshops for discussing these questions, and you directly communicate them with your tutor during the workshop on campus or online. If you're attending online workshops, you're more than welcome and encouraged to request one-to-one sessions if needed. For example, if you want to share your screen and show code to your tutor, that's perfectly fine. You can request that and the tutors should be able to set up a breakout room and help you. Okay, non-technical questions. So with non-technical questions, if there's any non-technical question that has not been answered earlier, through the assessment guideline, through the unit outline, this half lecture, if there's anything that has not been answered and you need clarification or if you have any special requests related to your circumstances, need adjustments, feel free to directly reach out to me. You can email me, the email address provided here. And the requirement is you need to always include this code, SIT112-STUDENT, in your subject line. This is to ensure that you get a timely reply. Otherwise, your email might get lost in my really busy mailbox. Or I might see it with delay and you might get a really late reply. So if you want a timely reply, make sure you always include this code in your subject line. Also, give two working days for response. If you didn't get a reply within two working days, feel free to follow up with me. Another point is that please do not use Microsoft Teams for contacting the unit chair. Use email instead. We are using email for communicating any issues and matters with the unit chair, that is myself.

29:54

Continuing on communication, we're going to communicate any important matters r



related to the unit. That means any issues, any updates, any changes related to the tasks, lectures, workshops, any news will be directly communicated to you via announcements. So you are expected to keep yourself up to date with the latest announcements on the CloudDeakin website. So announcements is the primary way that I will communicate any issues related to or any matters related to the unit with the students. Discussions can take place during the workshops. That's the primary place for discussions, but you can also continue the discussions beyond workshops and to facilitate that there is a student forum and some discussion topics already listed there so you can continue the discussion on these topics. You can have general discussions or talk about how you can use ChatGPT for data science and AI and you can share any interesting ways of using ChatGPT, how to for example generate code or how to check the correctness of your solutions for the tasks and different things, how you can find issues in the code. Also you can discuss specifically about tasks under pass tasks topic here or credit tasks or distinction and high distinction tasks.

31:34

So there are some best practices in this unit for you to achieve the best results. You can follow them. So work on your tasks as seriously as possible. Do not procrastinate. Demonstrate your work in the workshops to your tutors, seek feedback as early as possible, and utilize ChatGPT as we mentioned before. Also make sure your target grades are set realistically and feel free to communicate any obstacle related to the unit with your tutors and myself. Be respectful to the tutors and remember that they're at your side, they're there to help you achieve your target grade. If something stresses you about the unit, please feel free to communicate that. But before that, make sure that you try to calm down, write down the issue, make sure you fully understood it, refer back to the assessment guideline and other unit material, make sure that your concern is not addressed there. In many cases it is addressed, and if you need further help feel free to communicate your concern with your teachers and myself and we would be happy to help you. Also give your teaching team, your teacher, or myself a couple of days to come up with an actionable solution for your concern. And if you didn't hear from them, feel free to follow up.

03:20

Can something be done differently in this unit? This is a common question that you might get across different units, especially in large units, where there are more people



e with different preferences. We celebrate these differences, we celebrate diversity in this unit and recognize your preferences. We also welcome feedback for future enhancements. However, we will not be able to incorporate preferences into the running version of the unit, that means the current version. That is to avoid confusion and to ensure units are coherent and integrity. There's some references that you can use in this unit, but you're not required to. If you'd like to delve deeper into the topics that we're covering in this unit, you can use the references listed here. And of course, there are many other references available online. You can always consult with your GPT and seek more information or ask for other references. And if you need to have more exercise and improve your Python skills, there are some websites that you can use. Kaggle is one of the good ones. You can go there and there are plenty of data science tasks there. You can work on and can gradually work towards more sophisticated tasks. And of course there are plenty of other resources that are available online, so do not limit yourself to what we have listed here and always try to look for any valuable resources that you can use. This is the end of part one of lecture week one and see you in part two.